Prodigy Anesthesia Quick Review

COPD/Emphysema

What two diseases does COPD include?	Chronic bronchitis and emphysema *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 188.*
What is a potential adverse side effect of diuretic therapy to treat cor pulmonale in patients with severe COPD?	The use of diuretics can produce a hypochloremic metabolic alkalosis from the loss of chloride that can depress the ventilatory drive and exacerbate CO2 retention. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 190.*
What is the difference between chronic bronchitis and emphysema?	Chronic bronchitis is characterized by obstruction of small airways and emphysema consists of enlargement of the air sacs, destruction of the lung parenchyma, loss of pulmonary tissue elasticity, and closure of the small airways. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 188.*
What are the primary symptoms associated with COPD?	Chronic productive cough and progressive exercise limitation. In advanced disease, orthopnea and dyspnea during activities of daily living may develop. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 188.*
What is asthmatic bronchitis?	Asthmatic bronchitis is the combination of chronic bronchitis with reversible bronchospasm. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 188-189.*
How are pulmonary function tests affected by COPD?	The FEV1/FVC is decreased, but the FEV25-75 is decreased the most. The total lung capacity (TLC), residual volume (RV), and functional residual capacity (FRC) are all normal or increased. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 190.*
What chest x-ray findings are consistent with COPD?	Often, the chest x-ray doesn't reveal many changes, but if the disease is severe enough, there may be hyperlucency due to vascular deficiency in the lung periphery. Hyperinflation of the lungs may be seen as the diaphragm loses its curved appearance and becomes flat. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 189.*
How does the hematocrit differ between patients with chronic bronchitis and emphysema?	The hematocrit is elevated in patients with chronic bronchitis and normal in patients with emphysema. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 189.*

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How does smoking cessation affect the progression of COPD?	Smoking cessation can diminish or even completely eradicate the symptoms of chronic bronchitis and halts the progressive decline in lung function seen in patients who continue to smoke. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 190.*
What is the most effective therapy for treating increased pulmonary vascular resistance and erythrocytosis seen with COPD?	Chronic oxygen therapy, which is indicated for a PaO2 less than 55 mmHg or a hematocrit greater than 55%. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 190.*
Gas exchange occurs in the airways across which type of cells?	The mucosa in the airway gradually transitions from ciliated columnar epithelium to cuboidal epithelium and finally to flat alveolar epithelium where gas exchange occurs. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 489.*
What pulmonary function tests are indicative of a high risk for pulmonary complications from surgery?	FEV1 < 70% of predicted or FEV1/FVC ratio < 65% *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 195.*
What are the preoperative strategies that should be employed to reduce the risk of postoperative pulmonary complications in COPD patients?	Cease smoking for at least 6 weeks prior to surgery, treat any existing airflow obstruction, treat respiratory infections with antibiotics, teach the patient lung expansion techniques to use postoperatively. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 192-193.*
How is the duration of surgery related to the development of postoperative pulmonary complications in COPD patients?	Surgeries lasting 2.5 hours or more are associated with a higher risk of pulmonary complications in these patients. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 191.*
What is the role of regional anesthesia for patients with COPD undergoing surgery?	The use of regional anesthesia as the primary anesthetic is associated with a reduced risk of pulmonary complications when compared to general anesthesia and it should be employed whenever possible. The use of regional anesthesia for postoperative analgesia is also helpful because the increased pain control promotes greater lung expansion and earlier extubation. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 193.*

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What ventilator settings are most useful for patients with severe COPD who are undergoing general anesthesia?	Tidal volumes of 6-8 mL/kg, slower respiratory rates (6-10 breaths/min) and a slow inspiratory times minimize turbulent airflow and optimize ventilation-perfusion mismatching. This technique is often as effective in maintaining oxygenation as PEEP and allows the avoidance of the cardiovascular side effects of the application of PEEP. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 194.*
What preoperative diagnostic findings are associated with the highest likelihood that the patient with COPD will need continued mechanical ventilatory support following surgery?	A preoperative FEV1/FVC ratio less than 0.5 or a preoperative PaCO2 greater than 50 are associated with a high likelihood for the need for continued ventilatory support following surgery. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 195.*

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